DAY 07 • STUDENT HANDOUT

Security Solution Status Check

Checking that security solutions are installed, operational and able to clean or delete threats

Qualification	Cyber Threat Monitoring Level I
Unit of Competency	CS-ICT251101 Monitor and Report Cyber Threats
Coverage	Element 2 Check status of third-party security solution: Performance Criteria 2.1–2.3; Range of Variables: Security solution, Action; Required Knowledge 2.1 Log and detection management, 2.2 Security solution usage and operations, 2.3 Knowledge in organization SOP
Duration	8 hours (1 training day)

Learning Objectives

At the end of this session, you should be able to:

1. Explain why a monitoring analyst checks the security solutions themselves, not only their alerts.
2. Check whether the required enterprise security solutions are installed on every host, and find coverage gaps (PC 2.1).
3. Check whether each security solution is operational against the SOP checks (PC 2.2).
4. Check whether a security solution can clean or delete a detected issue, from its policy and the threat type (PC 2.3).
5. Run the full status check on a Windows workstation with PowerShell and report what fails.

1. Why Check the Security Solutions?

Every alert you handled on Days 5 and 6 came from a security solution. If that solution is missing, stopped, out of date or not allowed to act, there is no alert at all, and a quiet console looks exactly like a safe network. Element 2 of the Competency Standards asks three questions about every security solution. In the lab you follow SOC-SOP-03 Security Solution Status Check (~/day7/soc_status_sop.txt).

1. Is it installed? (PC 2.1)
2. Is it operational? (PC 2.2)
3. Can it clean or delete the issue? (PC 2.3)

The same three questions apply to every security solution in the Range of Variables: anti-virus, common firewall, threat intelligence, TDR, WAF, IDS, IPS, DLP, OT/ICS/SCADA security, IoT security, EDR/MDR, SIEM, SOAR and XDR. Today we practise on the three that sit on every computer: anti-virus (Microsoft Defender), EDR (CrowdStrike Falcon) and the SIEM agent (Wazuh).

2. Is It Installed? (PC 2.1)

The SOP says which enterprise security solutions every kind of host must have. Compare the organization's list of computers (the endpoint inventory) with what the security console reports.

Host Type	AV (Defender)	EDR (Falcon)	SIEM Agent (Wazuh)
Windows server or workstation	required	required	required
Linux server	—	required	required
Tier 3 test PC (isolated VLAN)	required	not required	required

> **NOTE — The host the console has never seen**
> A computer that was never enrolled does not appear in the console as "Not installed": it does not appear at all. Always start from the endpoint inventory and look for every host in the console report. Counting the lines of both files (wc -l) is a quick first check.

3. Is It Operational? (PC 2.2)

Installed is not the same as working. A solution is operational only when every check passes:

Check (SOC-SOP-03)	Passes When	Where You See It
B1 Service	Running	Get-Service
B2 Real-time protection	On (anti-virus)	Get-MpComputerStatus
B3 Check-in	No more than 60 minutes ago	Console health export; agent log
B4 Content	Signatures dated within 3 days	AntivirusSignatureLastUpdated
B5 Version	At or above the minimum supported version	Console health export
B6 Tamper protection	On	IsTamperProtected

> **TIP — "Running" does not mean "working"**
> A Wazuh agent service can be Running while the agent has lost its connection to the SIEM manager: nothing reaches the SOC. Read the last lines of its log (ossec.log): "Keep alive acknowledged" means connected; "Lost connection with manager" means the SOC is blind on that computer.

4. Can It Clean or Delete the Issue? (PC 2.3)

A solution can detect a threat and still be unable to remove it, because of its policy or the kind of threat. Microsoft Defender's policy (Get-MpPreference) sets a default action for each severity:

Action Code	Meaning
1	Clean
2	Quarantine
3	Remove
6	Allow

Action Code	Meaning
8	User defined
9	No action
10	Block

Rule	Meaning
C1 Action code	Can clean or delete only with action 1, 2, 3 or 10 for the threat's severity
C2 Exclusion	A folder in ExclusionPath is not scanned; user folders (Downloads, Desktop, Temp, Documents) must never be excluded
C3 Detect only	EDR in "Detect only" mode raises alerts but cannot stop or remove anything; required mode is "Prevent"
C4 Re-image	Ransomware, rootkits and boot-sector infections: the host is rebuilt by IT, even if the file was deleted

Actions in the Range of Variables

Action	Meaning
Failed	The solution tried and could not act: report it
Clean	Malicious code removed, file kept
Delete	File removed
Quarantine	File moved to a safe, locked place
Blocked	Stopped before it could run or connect
Re-image	Host rebuilt from a clean image

5. What to Report

- A missing required solution or a host never enrolled (coverage gap)
- A failed operational check, with the check code (B1–B6)
- A policy that cannot clean: the action code, the exclusion or the detect-only mode
- A detection that needs a re-image
- Report to the Shift Lead. Level I does not change policies or uninstall anything.

Key Terms

Term	Meaning
Enterprise security solution	Security software managed centrally for the whole organization
Agent / sensor	The small program on each computer that reports to the console
Coverage gap	A host without a required security solution

Term	Meaning
Check-in / keep-alive	The regular "I am alive" message from an agent to its console or manager
Tamper protection	A setting that stops users and malware from switching security off
Exclusion	A folder, file or process the solution is told to ignore
Detect only mode	The solution watches and alerts but never blocks or removes
Re-image	Wiping a computer and reinstalling it from a known clean image

References

- TESDA Competency Standards — Cyber Threat Monitoring Level I (ICT Sector), unit CS-ICT251101, Element 2 and Range of Variables 5–6.
- Microsoft Learn — Microsoft Defender Antivirus: Get-MpComputerStatus and Get-MpPreference (threat default actions, exclusions).
- Wazuh documentation — agent connection and keep-alive (ossec.log).
- NIST SP 800-137, Information Security Continuous Monitoring (ISCM) for Federal Information Systems and Organizations.
- NIST SP 800-83 Rev. 1, Guide to Malware Incident Prevention and Handling for Desktops and Laptops.